DATA SCIENCE MINI PROJECT

Paddy Crop Data Analysis and Prediction

PROJECT DOCUMENTATION

Prepared by	Harishraj R
Program / Batch	Data Science • DS-ANB-04
Project type	Supervised Machine Learning – Regression
Prediction target	Paddy yield (in Kg)

An end-to-end exploration of paddy cultivation data, feature preparation, regression modelling, cross-validation and model evaluation.

Executive Summary

This project examines a paddy cultivation dataset and develops supervised regression models to estimate paddy yield in kilograms. The notebook covers dataset inspection, exploratory data analysis, duplicate removal, categorical encoding, feature scaling, feature selection, train/test splitting, five-fold cross-validation, randomized hyperparameter tuning and comparison of three regression algorithms.

Project item	Notebook result
Original dataset	2,789 rows × 45 columns
After duplicate removal	2,338 rows × 45 columns
After one-hot encoding	2,338 rows × 64 columns
Selected input features	32
Train / test split	1,753 / 585 (75% / 25%)
Best test-set model	Gradient Boosting Regressor
Test-set R^2	0.989897

Main result

Gradient Boosting also achieved the lowest best cross-validation RMSE (820.553 kg), followed by Random Forest (827.276 kg) and Decision Tree (835.371 kg). It was selected as the strongest candidate by both the cross-validation result and the test-set comparison.

Important interpretation note: the notebook fits scaling and SelectKBest feature selection before the train/test split. This can allow information from the eventual test set to influence preprocessing and feature selection. The reported scores should therefore be treated as preliminary until the workflow is refitted inside a training-only pipeline and checked on independent farm, season or location data.

1. Project Overview

Agriculture plays an important role in food production. This project uses paddy cultivation data to explore agricultural patterns and investigate how machine-learning methods can support the estimation of paddy yield.

Problem statement

The project aims to identify useful patterns in paddy-related variables and build regression models that estimate the continuous target variable, **Paddy yield(in Kg)**, from cultivation and environmental features.

Objectives

- Understand the dataset structure, data types, descriptive statistics and missing values.
- Explore distributions and relationships using visualizations.
- Remove duplicate rows and encode selected categorical columns.
- Scale numerical columns and select a subset of useful features.
- Train and tune regression models using five-fold cross-validation.
- Compare model performance using MAE, RMSE and R^2.

Project classification

Item	Description
Domain	Agriculture / Data Science
Learning task	Supervised learning
Problem type	Regression
Target variable	Paddy yield(in Kg)
Tools	Python, NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn

2. Dataset Description

The notebook loads the CSV dataset from a GitHub raw-content URL. It contains 2,789 records and 45 original columns. The fields represent cultivation information, agricultural inputs, rainfall and irrigation-related measurements, temperature, wind, humidity and paddy yield.

Dataset source

UCI Machine Learning Repository: Paddy Dataset
CSV used in the notebook: GitHub raw CSV

Feature groups

- **Cultivation and farm details:** Hectares, Agriblock, Variety, Soil Types, Nursery and seed rate.
- **Agricultural inputs:** fertilizer, weed-control, pesticide and micronutrient measurements.
- **Weather and environment:** rainfall, irrigation, temperature, wind speed, wind direction and relative humidity across growth periods.
- **Output:** Paddy yield(in Kg), used as the continuous prediction target.

Dataset structure

Stage	Rows	Columns
Original data	2,789	45
After duplicate removal	2,338	45
After one-hot encoding	2,338	64

The notebook's missing-value inspection reports zero missing values for all 45 original columns. It identifies and removes 451 duplicate rows, reducing the dataset from 2,789 to 2,338 records.

3. Exploratory Data Analysis

Exploratory data analysis was performed to understand the target distribution and examine how paddy production varies across selected agricultural and environmental variables. The following figures are the original visualization outputs embedded in the supplied notebook.

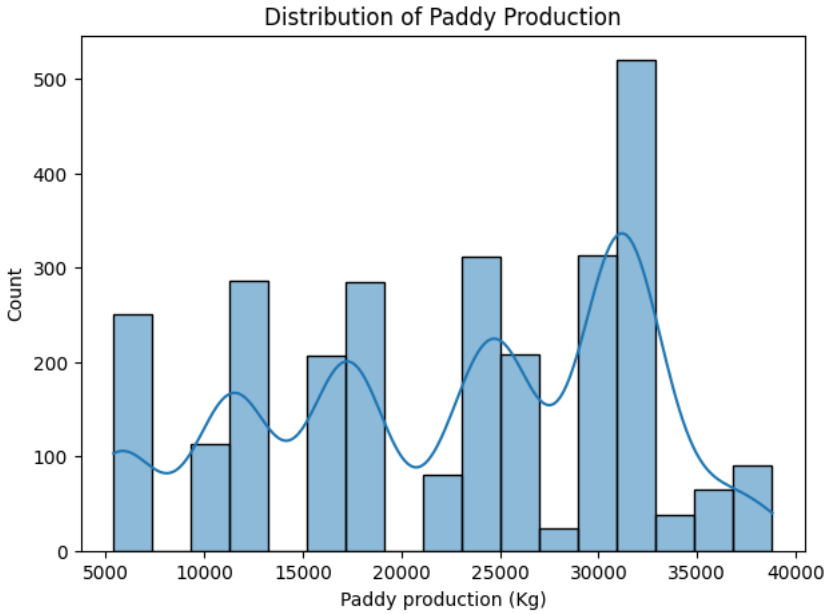


Figure 1. Distribution of paddy production (kg). The histogram and density curve show multiple visible clusters across the production range.

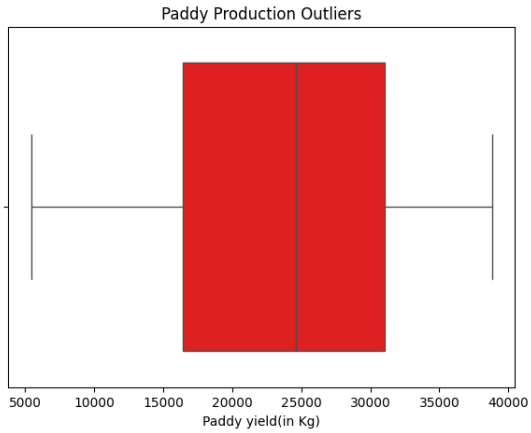


Figure 2. Boxplot of paddy production, used to inspect the spread and potential extreme values.

3. Exploratory Data Analysis (continued)

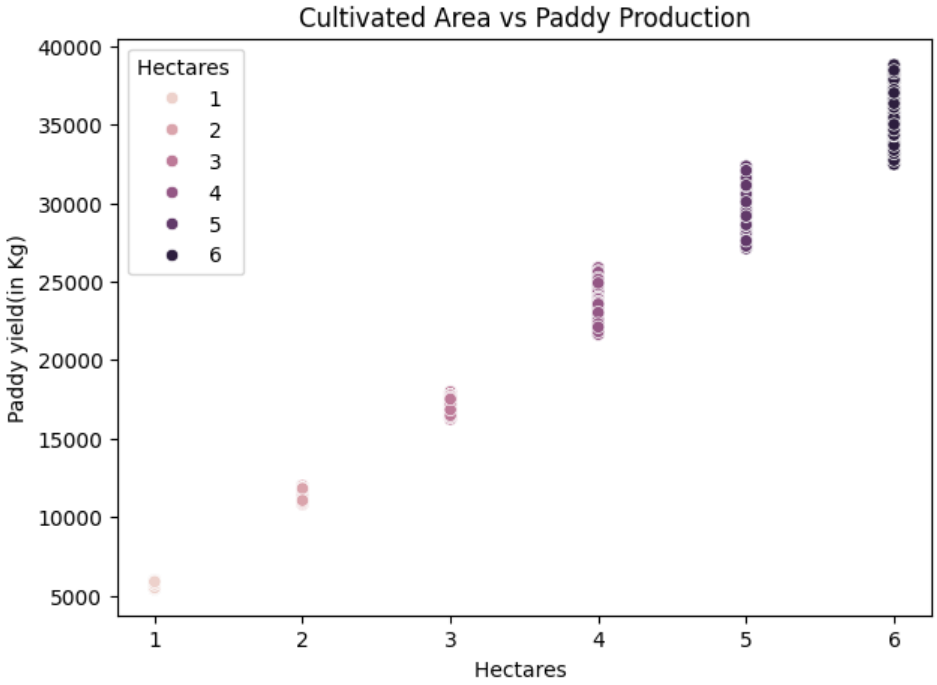


Figure 3. Cultivated area versus paddy production. The plotted values rise in groups as recorded hectares increase.

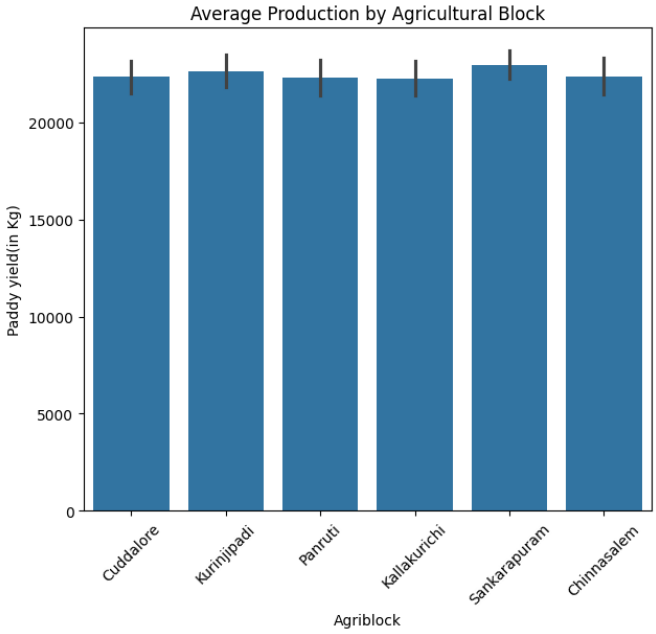


Figure 4. Average production by agricultural block. The plotted block averages are relatively close.

3. Exploratory Data Analysis (continued)

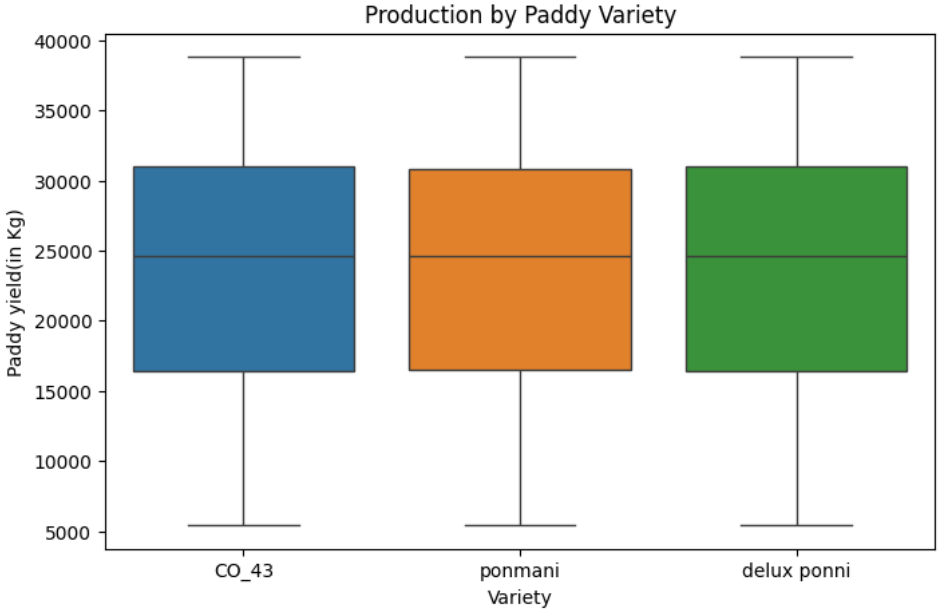


Figure 5. Paddy production by variety. The displayed distributions and medians appear broadly similar.

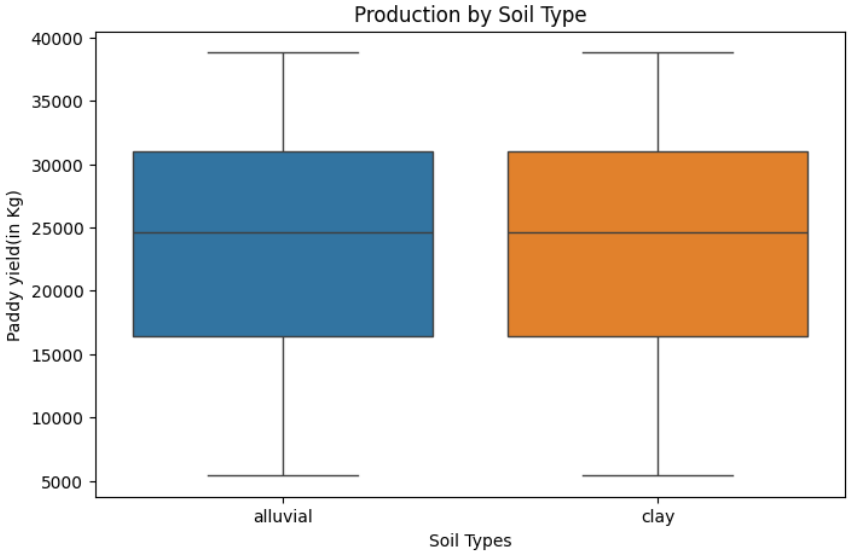


Figure 6. Paddy production by soil type. The two plotted distributions appear similar in median and spread.

3. Exploratory Data Analysis (continued)

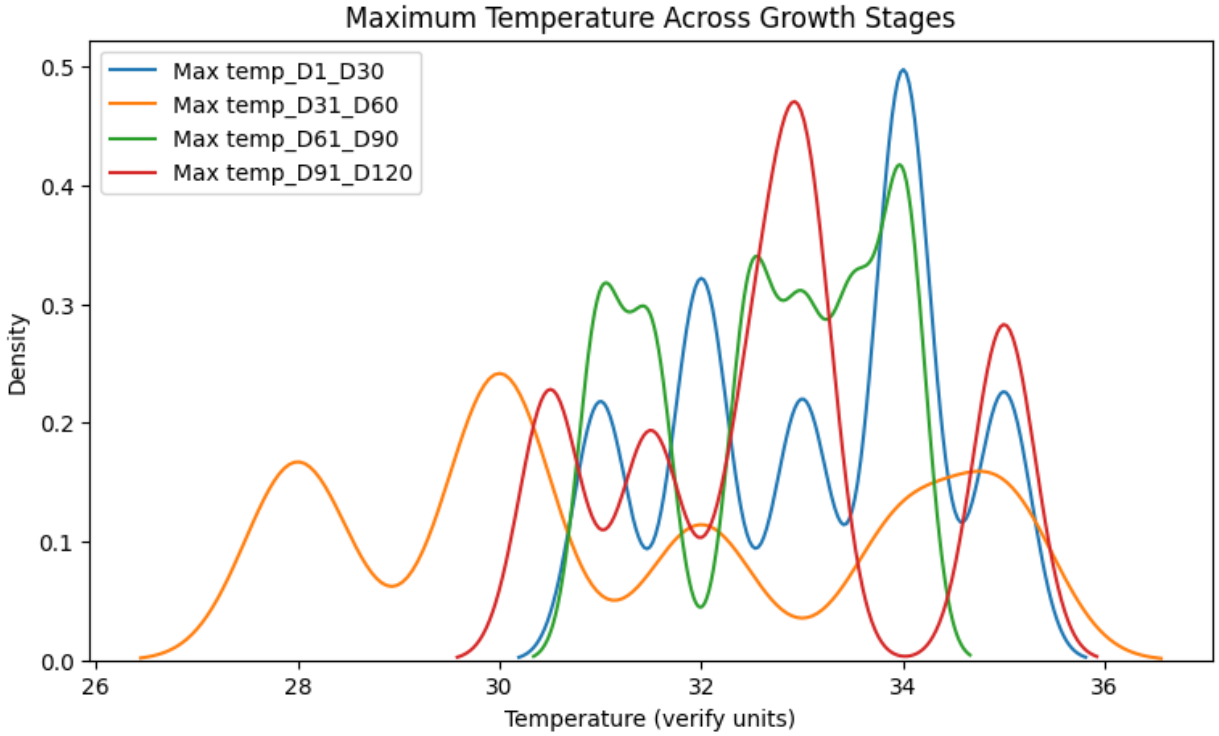


Figure 7. Maximum temperature distributions across four growth stages. Confirm measurement units and field definitions before drawing agronomic conclusions.

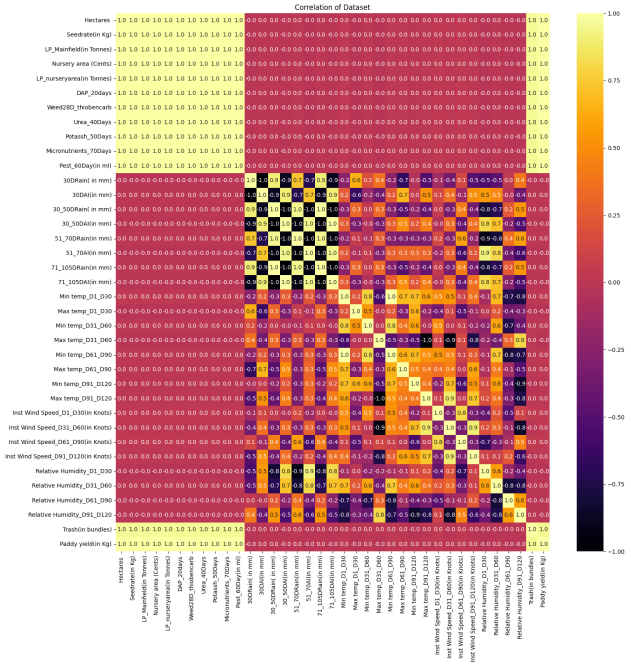


Figure 8. Correlation heatmap of numeric variables before categorical encoding. Several variables show very strong correlations.

3. Exploratory Data Analysis (continued)

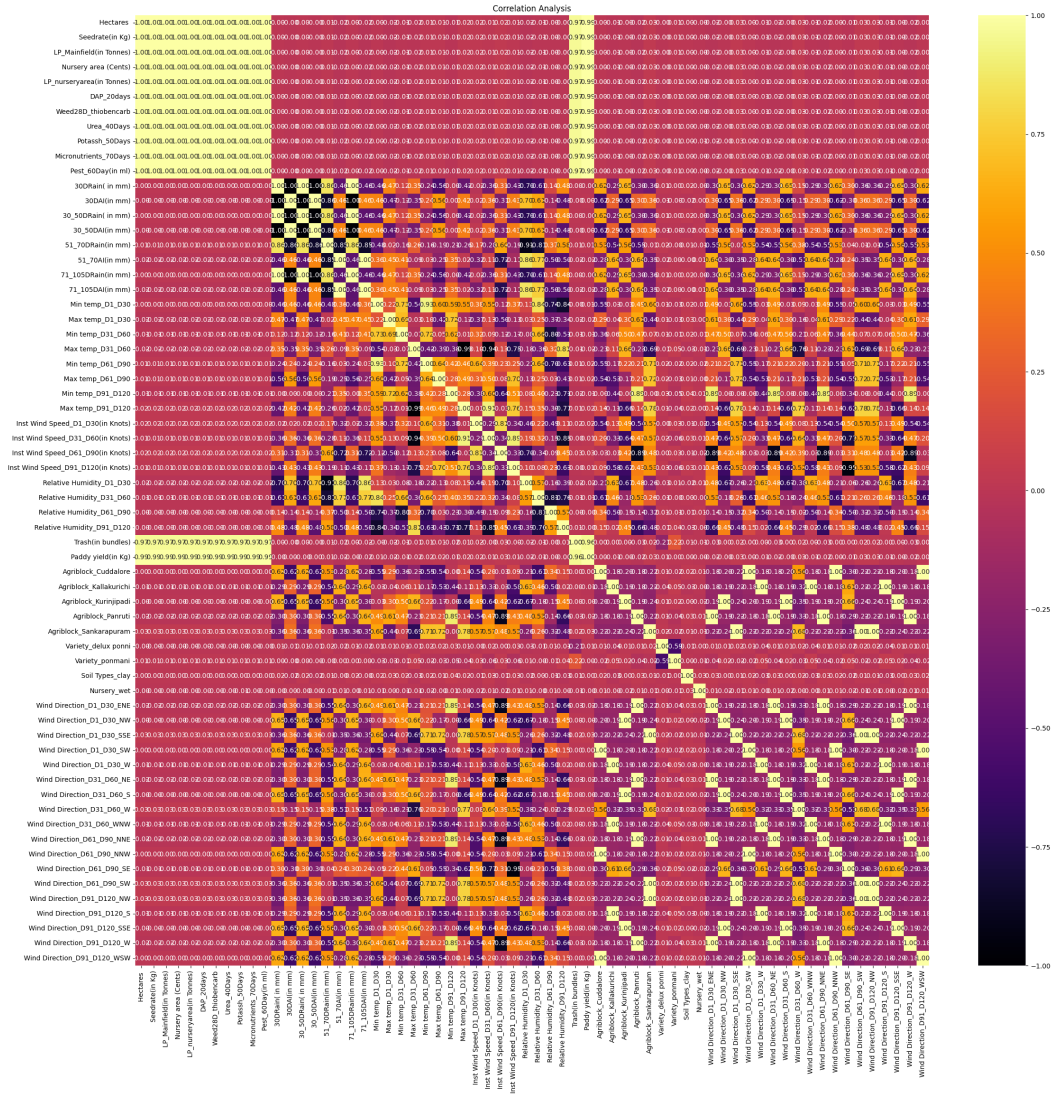


Figure 9. Correlation heatmap after one-hot encoding. The matrix includes encoded categorical features and numeric variables.

EDA observation

The notebook visuals suggest a strong relationship between cultivated area and production, while the correlation heatmaps show very high correlations among several input variables. These relationships should be checked for duplicated information or features that may be mathematically derived from the target.

4. Data Preprocessing and Feature Engineering

The notebook performs the following data preparation operations before model training:

Step	Implementation recorded in notebook
Missing-value inspection	data.isnull().sum(); all original columns show zero missing values.
Duplicate handling	data.drop_duplicates(); 451 rows removed.
Categorical encoding	pd.get_dummies() applied to Agriblock, Variety, Soil Types, Nursery and four wind-direction columns, with drop_firs
Feature scaling	StandardScaler applied to the numerical columns in df_encoded.
Feature selection	SelectKBest with f_regression; k=32 features selected.
Train/test split	train_test_split(test_size=0.25, random_state=42).

Feature selection

SelectKBest with the ANOVA F-test scoring function (f_regression) selected 32 input features. The selected set includes cultivation inputs, selected weather measurements, selected agricultural-block indicators and wind-direction indicators.

Train/test partition

The notebook uses a 75% / 25% split with random_state=42. This results in 1,753 training rows and 585 test rows.

Implementation note

In the supplied notebook, StandardScaler is fitted before the train/test split, and SelectKBest is also fitted before the split. To avoid information leakage, these transformations should be fitted only on the training data, preferably as steps inside a Scikit-learn Pipeline evaluated within cross-validation.

5. Regression Models and Validation

The notebook compares three supervised regression algorithms. The target is continuous paddy yield in kilograms.

Model	Role in the experiment
Decision Tree Regressor	Learns decision rules from the input features.
Random Forest Regressor	Combines multiple decision trees.
Gradient Boosting Regressor	Builds trees sequentially to improve on previous prediction errors.

Cross-validation and tuning

The notebook uses KFold with five splits, shuffling enabled and random_state=42. RandomizedSearchCV evaluates 20 sampled hyperparameter combinations for each model using negative root mean squared error as the scoring function. The tuning loop wraps each regressor in a Pipeline with a median SimpleImputer followed by the model.

Best cross-validation RMSE

Model	Best CV RMSE (kg)
Decision Tree	835.371
Random Forest	827.276
Gradient Boosting	820.553

Gradient Boosting has the lowest best cross-validation RMSE in the supplied notebook output (820.553 kg), followed by Random Forest (827.276 kg) and Decision Tree (835.371 kg).

Recorded Gradient Boosting settings

Gradient Boosting: learning_rate ≈ 0.0812; max_depth = 2; min_samples_leaf = 7; min_samples_split = 12; n_estimators = 121; subsample ≈ 0.8796.

6. Model Evaluation

The tuned models are evaluated on the held-out test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). Lower MAE and RMSE indicate smaller prediction errors; an R^2 closer to 1 indicates a stronger fit on the evaluated data.

Model	Test MAE (kg)	Test RMSE (kg)	R^2 Score
Gradient Boosting	665.375	922.675	0.989897
Random Forest	668.226	926.558	0.989812
Decision Tree	670.749	931.203	0.989709

Best test-set result

Gradient Boosting achieved the lowest test MAE (665.375 kg), the lowest test RMSE (922.675 kg) and the highest test R^2 (0.989897). It is therefore the top-performing model on the notebook's test-set comparison.

Cross-validation and test-set interpretation

Gradient Boosting also achieved the lowest best CV RMSE in the notebook output. The close test-set R^2 values (approximately 0.99) indicate a strong fit on this particular split. They do not, by themselves, demonstrate that the model will generalize to new farms, locations or growing seasons.

Overfitting / underfitting

The notebook's written evaluation correctly notes that training-set metrics are needed to confirm overfitting or underfitting. Test-set scores alone are insufficient to establish either condition.

7. Limitations and Recommended Next Steps

The following points are important before interpreting the model as a reliable real-world paddy-yield forecasting system:

- **Review duplicate records:** investigate whether the 451 repeated rows are true duplicates or repeated but valid observations.
- **Audit target leakage:** inspect features that may directly encode or be mathematically linked to production or cultivated area.
- **Prevent preprocessing leakage:** place scaling and feature selection inside a Pipeline and fit them only within training folds.
- **Use independent validation:** test on a separate season, farm or geographic block to estimate real-world generalization.
- **Document measurement units:** confirm the units and definitions of temperature, rainfall and irrigation fields.
- **Deployment:** after validation, save the complete fitted preprocessing-and-model pipeline before building an API or other deployment workflow.

8. Conclusion

This project demonstrates an end-to-end supervised regression workflow for paddy-yield analysis. It includes dataset inspection, exploratory visualization, duplicate removal, categorical encoding, scaling, feature selection, train/test splitting, five-fold cross-validation and randomized hyperparameter tuning.

Based on the supplied notebook's test-set metrics, Gradient Boosting is the strongest of the three evaluated models, with MAE of 665.375 kg, RMSE of 922.675 kg and R^2 of 0.989897. These results are promising for the evaluated dataset split, but the preprocessing order and strong feature correlations require further investigation before claiming reliable performance on unseen agricultural data.

Reproducibility and source note

This documentation was prepared from the supplied Jupyter notebook and its embedded visualization outputs. Numerical model results in this document follow the notebook's recorded outputs. The separately supplied PDF report contains some different historical metric values; the notebook was treated as the primary source for model metrics. The analysis has not been independently re-run.

Dataset CSV: https://raw.githubusercontent.com/HARISHRAJ-R-07/Project-dataset-csv-files/refs/heads/main/paddydataset.csv